Does AI adoption enhance productivity without proportionately increasing employment in the formal sector? A comparative study of the USA and India.

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1. Abstract

This study examines whether the adoption of AI enhances productivity without proportionally increasing employment in the formal sector, comparing trends in the United States and India. There are plenty of examples of productivity improvements that are driven by AI. Generative AI tools have increased service productivity by 15%, while developers using AI assistants are completing tasks up to 56% faster. Unfortunately, we appear to be far behind our productivity gains when it comes to increasing employment. The U.S. has half of the existing jobs with some potential for exposure to AI, and even amid climbing wages for not only specialized roles, but the demand for routine jobs that are automated, to preclude the presumed demand for jobs. In India, growth in AI-related job postings and salaries has been rapid; however, it's unclear whether the AI and/or external market forces are suppressing the growth rate of non-AI jobs in India, as we see a statistically significant reduction in non-AI job postings within three years of AI job posting adoption. These trends capture consistency in the productivity growth following AI adoption, which isn't matched by job creation (particularly, mid and low skill work). Policymakers must develop large-scale, population-level reskilling, policy reform regarding the education of the community, and broaden inclusion regarding AI adoption to safeguard labor-market impacts.

2. Introduction

"An innovation as powerful as the steam engine." - McKinsey & Company, 2023

Artificial Intelligence is no longer a futuristic term or phrase; it is a tremendous advancement in the modern world that has revolutionized how companies operate and economies develop. Unlike past technologies, such as smartphones or the internet, AI does not just act as a library of information. As McKinsey & Company rightly states, this innovation is as mighty as the steam engine due to its ability to boost the labour productivity of workers and adapt, learn and reason.

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines.

AI is used by firms to reduce human error, optimize supply chains, and automate customer service. The enthusiasm for AI's potential in various fields, such as medical diagnosis, is matched with growing concern over its dangerous consequences, particularly for employment. In many industries, especially in the formal sector, AI does not just assist employees but replaces them entirely. This has widened the gap between rising productivity and static employment in the formal sector. The formal sector refers to the industries and firms recognized and regulated by the government.

This research paper investigates the question:

Does AI adoption enhance productivity without proportionately increasing employment in the Formal sector?

The answer to this question holds immense importance. In the United States of America (USA),businesses face perpetual skill shortages and a labour force with an aging population. However,in India, the challenge is different. There is a rapidly growing labour force and an economy thatis required to generate more than 10 million jobs each year.

Artificial Intelligence (AI) adoption in the formal sector refers to the integration of advanced computational tools, such as machine learning algorithms, natural language processing, and predictive analytics, in tax-compliant businesses and organizations. Businesses and organizations in the formal economy may aim to improve efficiency, automate digitized and repetitive activities, and support decision-making with data to produce measurable productivity improvements. Unlike informal sectors, formal enterprises generally possess the infrastructure, financial resources, and skilled workforce required to implement AI at scale.

The issue of whether productivity growth induced by AI will raise the number of jobs created commensurate with the jobs displaced is particularly important for this project as it contradicts a major theme observed when previous technological revolutions created new industries that ultimately created jobs in those industries. Unlike previous technologies, AI not only complements manual labor by automating a function, but AI can also automate forms of analytic cognition and thus make several cognitive and analytic decisions that could eliminate and replicate certain functionalities of worker roles. Therefore, it raises concerns around the loss of mid-skill and lower-skill worker jobs, while only benefiting a diminishing number of highly-skilled professionals. The dichotomy of the USA and India is particularly informative, given the difference in economic variables, labour market variables, and technological sophistication. It is vital to understand the impact of AI productivity observed in both countries relative to employment, wages, and other outcomes.

3. Literature Review

Theoretical Background

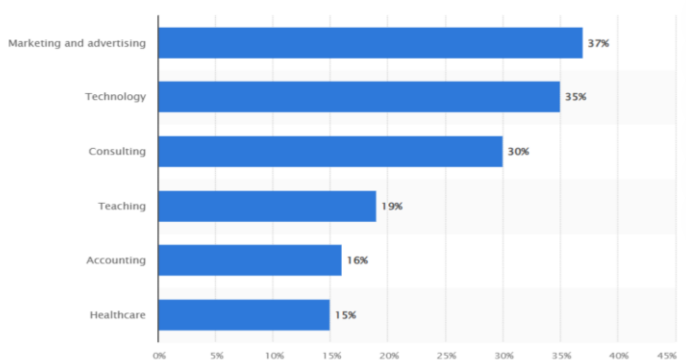
Much of the discussion about AI's influence on labor markets is framed within the automation–augmentation framework. Automation refers to the replacement of human labor in main, repetitive tasks. We may not be able to disassociate wage increases from productivity increases if an economy does not build new demand for labor, as a result of automation (Acemoglu & Restrepo, 2017). Augmentation suggests that AI may augment human capabilities, which may lead to productivity increases rather than reductions in work (Brynjolfsson & Hitt, 2022). Recent systematic reviews of different AI forms emphasize the need to contrast these effects, because not every sector/firm may be capable of augmentation versus total automation (Annual Reviews,2023).

Global and Regional Empirical Studies

Global reviews indicate that AI adoption has substantial labor market effects. The International Monetary Fund (2024) estimates that globally, nearly 40% of jobs face exposure to AI, which could substantially displace human labor (potentially in advanced economies, where there is the highest possibility of AI mimicking cognitive tasks). McKinsey & Company (2018) estimates that AI could generate considerable productivity increases, though there will be unequal gains across the relevant economic sectors and firms. According to PwC (2025), AI-exposed occupations have a fourfold productivity increase, and also a 56% wage premium for skilled workers; presumably in relation to products and technologies as opposed to services. However, employment growth remains modest.

In the United States, several studies have shown that early adopters did gain productivity quickly; however, they posted fewer non-AI roles within 3 years (IMF, 2024). By comparison, India's labor market was characterized by rapid growth in the number of AI jobs posted by 2012 to 2019, concentrated in IT and the finance sectors; however, it also displayed a measurable decrease in the number of non-AI-related vacancies, particularly in administrative and routine jobs (Ganuthula and Balaraman, 2025). Research distinguishes between "automation-AI," which tends to displace lower-skill workers, and "augmentation-AI," which creates new, specialized opportunities and wage premiums (Marguerit, 2025).

Rate of generative AI adoption in the workplace in the United States 2023, by industry



Source: Statista

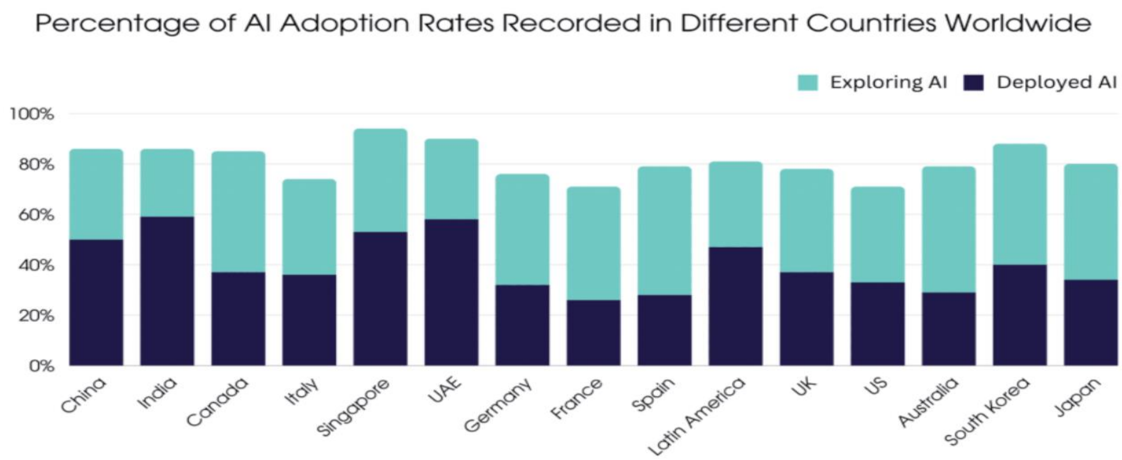
Findings from IMF, McKinsey, PwC, and HBR

IMF (2024) highlights that AI's benefits are uneven and require strong policy support to protect vulnerable groups. McKinsey & Company (2025) emphasizes that firms adopting AI earlier ("front-runners") achieve disproportionately higher productivity gains, widening competitive gaps. PwC (2025) demonstrates that even automatable jobs can see employment growth if AI is implemented as a complement rather than a substitute. Harvard Business Review (2021) adds that productivity benefits do not diffuse automatically; leadership strategies and employee training are critical to achieving broader labor-market gains.

Gaps in the Literature

While many studies examine AI impacts in developed economies, there are fewer studies interested in the impacts of AI in developing countries. There has been comparatively little research into how countries with different levels of readiness for new technologies will be impacted by AI adoption, and how countries with different levels of readiness will see categories of employment, wages, and skill demand change over time. More notably, we cannot find direct studies that look at to which demographic pressures, labor informality, and skill mismatches influence AI-induced employment mechanisms in the U.S.-India context, specific to people with disabilities.

4. AI Adoption Trends



Source: DemandSage

United States

The adoption of AI in the US has exploded in the time period following COVID-19. A PwC survey revealed that 52% of US companies utilized more AI during 2020. Demand for AI-related jobs also increased substantially, algorithm and automation-related jobs increased by 28% or more year-over-year (Harvard Business Review, 2021). Often, jobs in these fields also carry strong wage premiums, as workers have received 13% to 17% higher wages than their non-AI skilled counterparts coming out of the pandemic (International Monetary Fund, 2024).

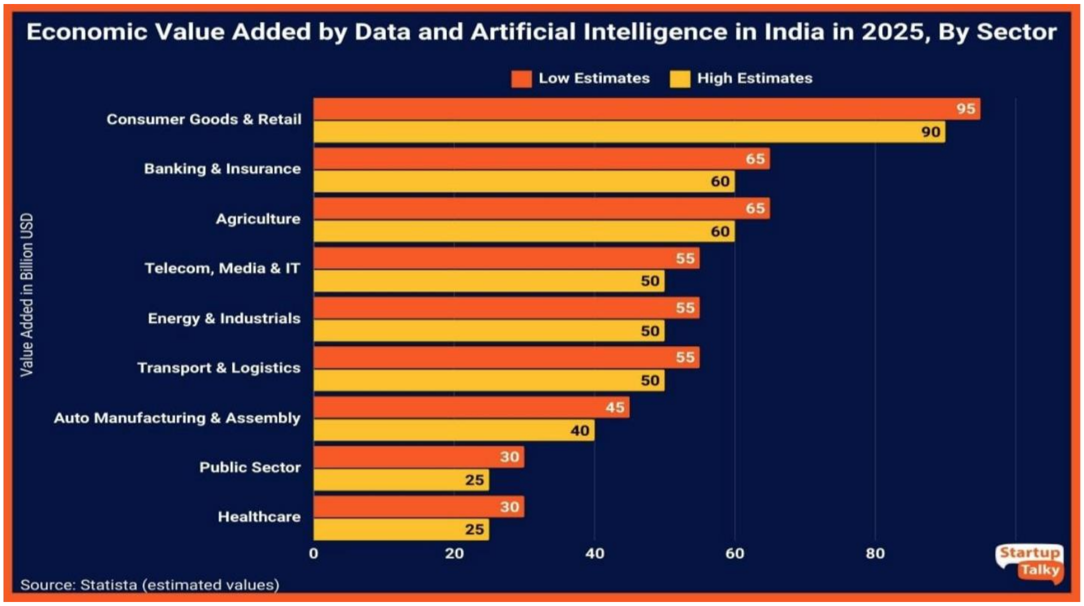
Different industries demonstrate disparate adoptions of AI, but some of the fastest adoptions have been in the finance, supply chain management ,and technical support fields. For instance, in banking, firms like the Bank of America have incorporated "responsible AI" to automate aspects of customer service while remaining compliant with regulators (Harvard Business Review, 2021). Also, the current supply chain crisis prompted companies to turn to AI analytics to develop predictive demand models, and digital product companies seeking to increase support for watered-down AI services, such as chatbots that could handle problems not too complex.

India

India experienced an exponential increase in AI-related activity, particularly between 2012 and 2019, where the number of online job postings that required AI skills increased rapidly enough to surpass not only the U.S. but also Australia, Canada, and the U.K. in 2019 (International Monetary Fund, 2024). The demand was highly concentrated in metropolitan areas, such as Mumbai, Delhi, Chennai, and Pune, which were driven by local diffusion effects that influence the uptake of AI across the entire sector.

The IT services sector accounts for the most occurrences of AI usage and job creation, which is a significant contribution to AI-related employment and training. Workers in India with any AI skills receive a 13 percent to 17 percent wage premium typical of U.S.-based trends (IMF, 2024). Despite this growth in demand for AI skills, overall job absorption is still somewhat limited given: Growing demand for workers with AI related skills leading to the rapid decline in non-AI postings; In studying the application of AI across firms, Ganuthula and Balaraman (2025) observed 1 year of decline around the time of implementation, and by year three there were observed declines of approximately 1% of postings, fewer non-AI postings.

Further, the projected opportunity for over 200 million youth entering India's workforce by 2030 will create further limitations if India can't transform faster technological effects of AI use into quality uploads of employment opportunities.

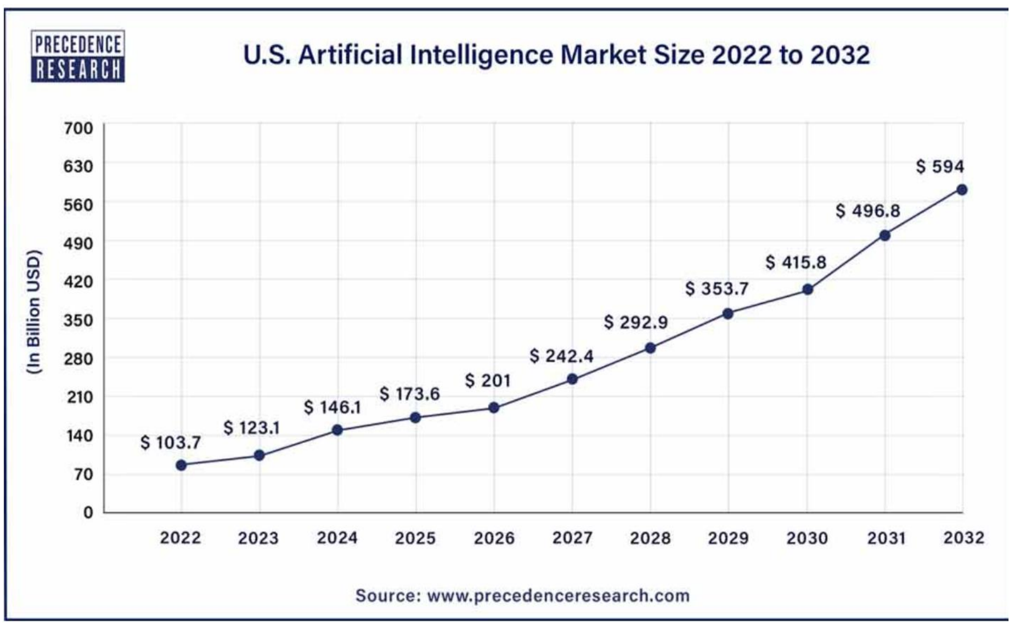


5. Impact on Productivity

The adoption of AI has become a crucial driver of increased productivity by providing increased efficiency, improved quality, and lower costs. AI allows the automation and streamlining of repetitive and routine processes and the ability to leverage data and make better informed decisions, enhancing the best use of resources and improving accuracy with outputs. Unlike previous digital tools, AI can process large volumes of data, find patterns, and predict trends, which makes it easier to reduce waste, enable quicker cycle times, and free up labor time.

United States

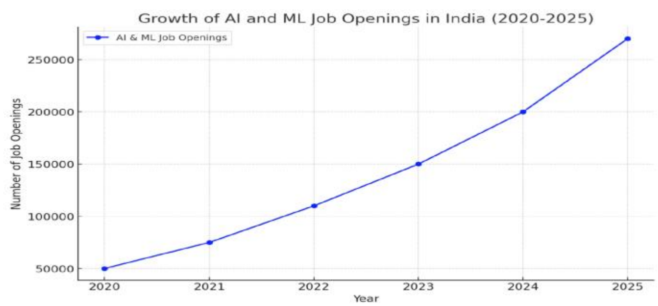
In the U.S., firms have recorded discernible productivity gains post-adoption of AI. Frito-Lay, for example, launched its online direct-to-consumer service, snacks.com, in 30 days, amidst the pandemic of COVID-19, consolidating five years of digital transformation plans into months. Frito-Lay was able to analyse consumer preferences with AI-driven analytics and adjust product offerings, which uses inventory more cost-effectively and at the same time increases sales (Harvard Business Review, 2021). Similarly, Bank of America employed "AI that's Responsible" technology and systems to reduce operational risks and significantly increase customer confidence. AI has been applied to predict disruptions and optimize logistics, enabling firms to maintain operations despite global shortages (McKinsey & Company, 2018).



India

In India, the economy's gains from productivity are easiest to identify within the IT services sector, where various AI tools support production and consumption processes by increasing productivity in software testing, cybersecurity, back-office functions, and customer service functions. In finance, firms are utilizing machine learning tools for credit scoring, fraud detection, and personalized customer services, which are highly impactful because they increase efficiency and decrease default-related costs. Additionally, we see an emerging wave of AI-driven startups that use advanced analytics and other capabilities of AI to revolutionize established business models in the fields of healthcare, education, e-commerce, and others. However, interpreting productivity gains from AI has several caveats, as many of the resource and infrastructure differences dilute the potential gains of AI in urban settings, particularly in firms that have adopted more divergent industry standards.

6. Impact on Employment



Source: HaridwarUniversity

The adoption of AI has shown a very similar trajectory for its short-term impacts on labor. When companies first make the decision to adopt an AI tool, the firms do not immediately reduce their non-AI job postings. In fact, it appears they experience a lag of some kind, rather than figuring out where to parse labor (International Monetary Fund, 2024). However, longitudinal data (one to three years) shows that companies that adopt AI, on average, post about 1% fewer non-AI job postings than similar firms that do not adopt AI (Ganuthula & Balaraman, 2025). This reflects that task substitution was achieved, albeit at a very slow pace ,rather than immediate job demise.

Occupational Effects

The impact on labor market demand varies dramatically by occupation. For example, managers and engineers, particularly in roles involving predictive analysis, data analysis, and optimization of processes, face the greatest demand as their intellectual and analytical functions are automated by AI tools (Marguerit, 2025). There is less change after AI adoption in sectors that involve manual labor work like construction jobs, where demand in AI-related job postings has shown little to no change; it is clear that heavy labor types of jobs are less susceptible to near-term automation.

Wages and Skills

For people who have AI skills, the labor market incentives are high. Workers with AI skills earn a 13–17% premium reflected in a specialized education and technical expertise (IMF, 2024). However, while a premium exists, there are too many skill shortages, limited access to training, and opportunities are not evenly distributed. This was especially clear in developing economies like India, where a significant portion of the workforce exists outside of advanced digital ecosystems. There is a risk of exacerbating inequalities between professionals in high-skilled urban employment and the wider labor market.

7. Comparative Analysis: USA vs. India

Similarities

There is strong evidence in both the US and India that AI adoption is positively correlated with firm- and industry-level productivity. It is also interesting to note that the motivations for AI investment vary. AI implementation is most concentrated in industries involved in technology, for example, IT services, finance, and supply chain. The highest incidence of AI adoption exists in the technology sector, while industries that rely on labor have yet to adopt AI in any significant way.

Surveys have also indicated that employees in the US and India view AI as positively-more efficient. However, there remains a shortage of structured, large-scale training programs and therefore, which restricts access for the wider labor market to transition from low/semi-skilled work to AI-augmented roles(McKinsey & Company,2023)

Differences

The United States demonstrates a more widespread diffusion of AI adoption across a wide range of sectors of the economy, including healthcare, retail, logistics, and manufacturing, enabled by significant investments in digital infrastructure and business investment in AI. In contrast, India's AI growth is concentrated largely in urban centres and IT-dominant sectors. Despite significant and rapid increases in AI-related job postings overtaking the U.S. in 2019), India is more constrained by structural conditions related to job absorption, given the large and growing youth population, the prevalence of skills mismatches, and rigid labour market dynamics. Thus, productivity improvement stemming from AI risks growing faster than job creation, particularly outside of urban technology clusters.

8. Policy Implications

The evidence suggests that while AI adoption creates substantial productivity growth, it does not necessarily follow that employment grows proportionately. This evidence suggests that targeted policy responses are necessary to ensure that technology's dynamic nature supports functional aspects of inclusive economic and social change.

1. Workforce Reskilling and Education Reform

The U.S. and India need substantial investment in reskilling programs to prepare workers for

AI-augmented work. The worker demand for AI-literate employees, which comes with the associated salary premium, clearly suggests that workers with specialized skills are more rewarded (International Monetary Fund, 2024). Policies ought to encourage technical training, digital literacy, and continuous learning frameworks to avoid skill-based inequality.

2. Support for Job Transition and Safety Nets

The loss of non-AI job postings begins within three years of adoption. Governments will want to enhance unemployment benefits, wage insurance, and job placements to assist in transitional labour-market disruptions.

3. Encouraging AI Augmentation Over Full Automation

Policy frameworks can incentivize firms to use AI to augment rather than completely replace human labour. Tax incentives, grants, or regulatory advantages could promote “augmentation first” strategies that maintain employment while boosting productivity (Brynjolfsson & Hitt, 2022).

4. Bridging Regional and Sectoral Gaps

In India, the benefits of AI exploitation have been in urban information technology hubs, at the cost of those in rural areas or traditional occupations. Policymakers will need to ensure that infrastructure, as well as digital access, is expanded to mitigate a worsening urban-rural employment inequality. In the United States, the focus of policy should be similarly expanded beyond industries requiring intensive use of technology.

5. International Collaboration

As AI technology and policies are reshaping the labour market, it is important to collaborate on policies focused on common ethical standards, data use, and data sharing regulations to help curb further widening of the gap between developed and developing economies.

9. Conclusion

AI adoption is a game-changer that improves worker productivity through operational efficiencies, cost reductions, and new types of innovation. Evidence from both the US and India shows some common trends:

- Firms applying AI can see measurable increases in their output
- Workers who have skills in AI are compensated at a significantly higher rate.

However, the improvement in productivity has not coincided with an equivalent increase in employment levels. In fact, there tends to be a decline in non-AI job postings within three years of adoption: managerial and analytical jobs tend to have the sharpest declines. The US has the advantage of greater diffusion because AI supports existing jobs and sectors, whereas India has more structural issues to face, especially with rapid labor force growth, skill mismatches, and limited absorption for those jobs that do exist.

Future research may consider the long-run impacts of AI adoption on the labor market; for example, do new sectors offset unwanted job displacements? Which sectors are able to enhance their productivity with AI while retaining jobs? It would also be worth examining the success of any policy interventions, especially when it comes to large-scale upskilling initiatives or codesigned or incentivised approaches to self-learning. Finally, the issue of a sector-specific approach may lend itself to more informed discussions about industries that are better suited to linking productivity growth through AI while having sustainable employment opportunities.

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